

Laxmi Trust's Charitable
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – VIII

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A. Add and Interpret Trend Lines

1. Create Temperature vs Humidity Trend

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Connect to the file Indian_Climate_Dataset_2024_2025.csv.
3. Open a New Worksheet.
4. Rename the worksheet to Temperature vs Humidity Trends.

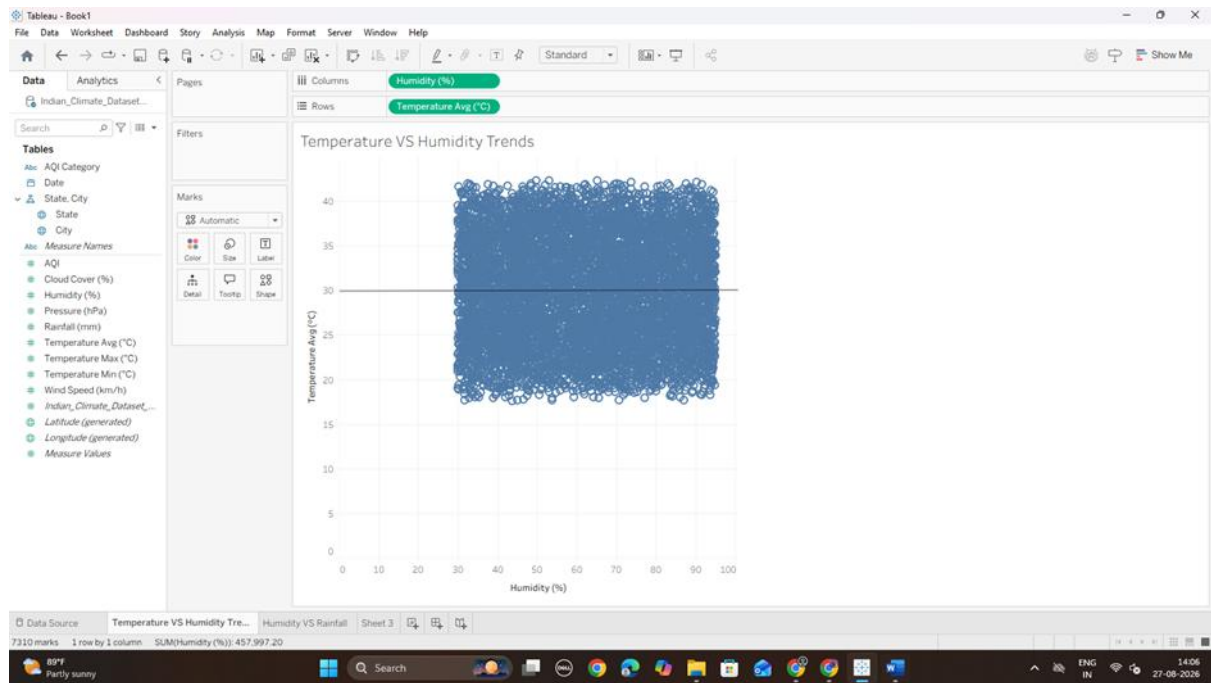
Step 2: Create the Chart

1. Drag Humidity (%) to the Columns shelf.
2. Drag Temperature_Avg (°C) to the Rows shelf.
3. Go to the Analysis menu.
4. Uncheck Aggregate Measures so that individual data points are displayed.

Step 3: Add Linear Trend Line

1. Click the Analytics pane on the left side.
2. Under Model, drag Trend Line onto the chart.
3. Drop it on Linear.
4. Hover over the trend line to view:
 - Linear equation
 - R² value
 - P-value

Result



A linear trend line is created to show the relationship between Humidity and Average Temperature.

2. Create an Exponential Trend Line

Step 1

1. Open a new worksheet.
2. Rename it Humidity vs Rainfall.

Step 2

1. Drag Humidity (%) to Columns.
2. Drag Rainfall (mm) to Rows.
3. Go to Analysis and uncheck Aggregate Measures.

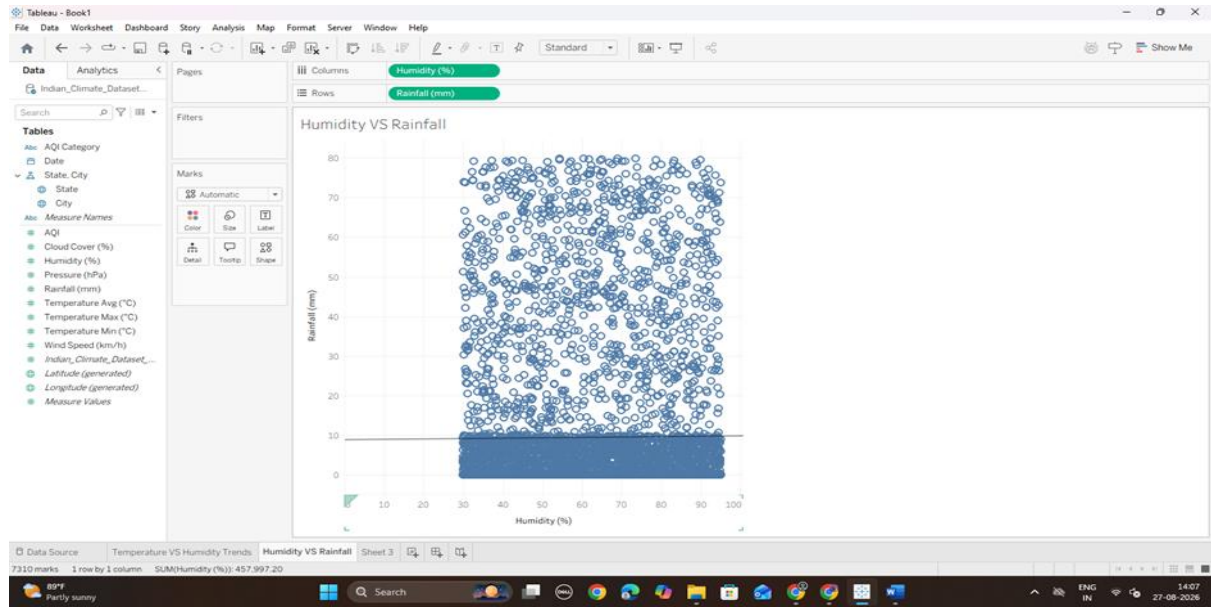
Step 3

1. Open the Analytics pane.
2. Drag Trend Line onto the chart.
3. Drop it on Exponential.

Alternative Method

- Right-click the chart.
- Select Trend Lines → Show Trend Lines → Edit Trend Lines.
- Change the model type to Exponential.

Result



The exponential model helps analyze non-linear relationships between humidity and rainfall.

B. Customize and Analyze Trend Models

1. Create Trend Lines by City

Step 1

Open the Temperature vs Humidity Trends worksheet.

Step 2

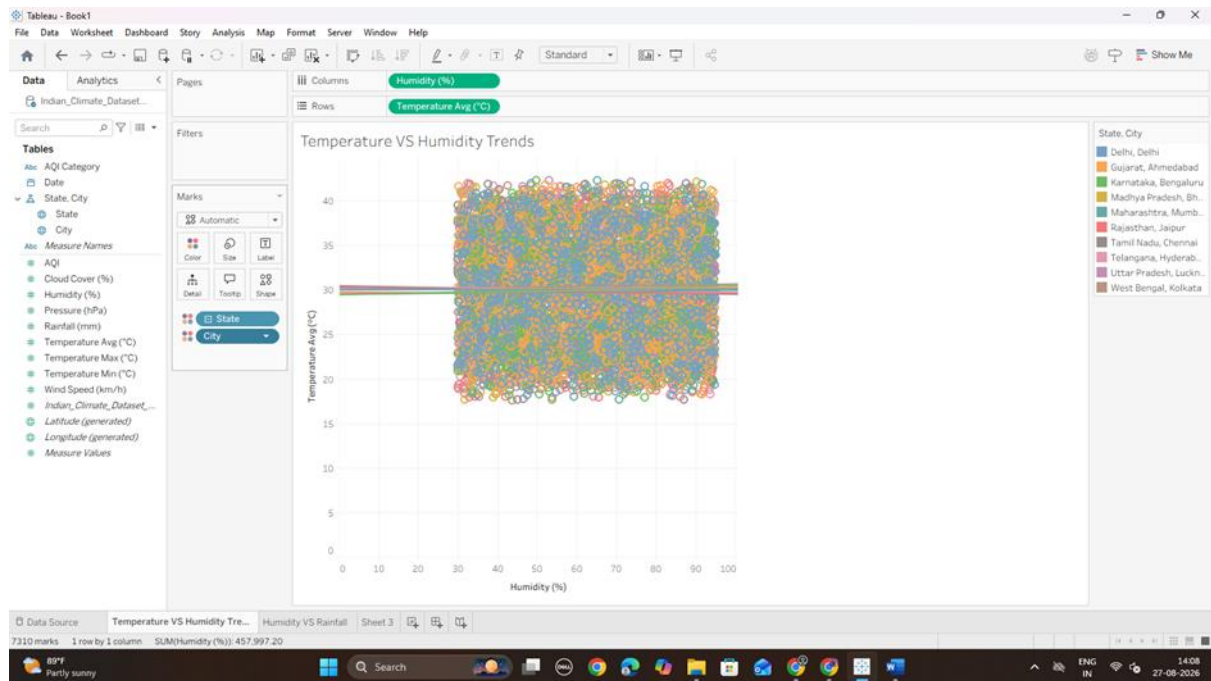
Drag City from the Data pane to Color on the Marks card.

Step 3

Observe the chart.

Tableau creates separate trend lines for different cities, allowing comparison of their climate patterns.

Result



Different cities can be compared using their individual trend lines.

2. Compare Different Model Types

Step 1

Right-click any trend line.

Step 2

Select Edit Trend Lines.

Step 3

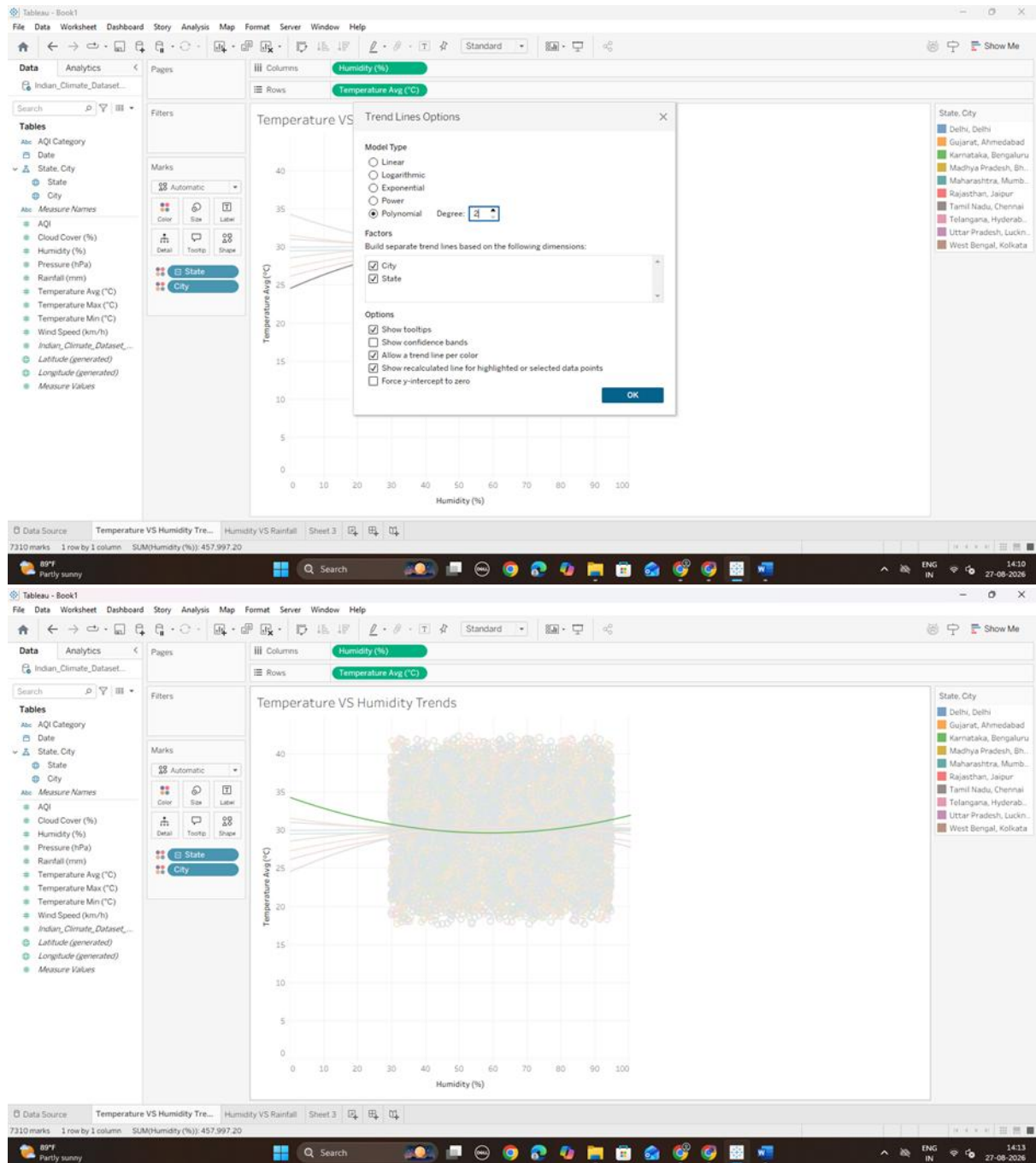
Experiment with different models:

- Linear
- Logarithmic
- Polynomial Degree 2
- Polynomial Degree 3

Step 4

Select Polynomial (Degree 2) and click OK.

Result



Different statistical models can be compared to determine which model best fits the data.

3. View Statistical Summary

Step 1

Right-click the chart or trend line.

Step 2

Select Describe Trend Line.

Step 3

Check:

- Formula equation
- R^2 value
- P-value

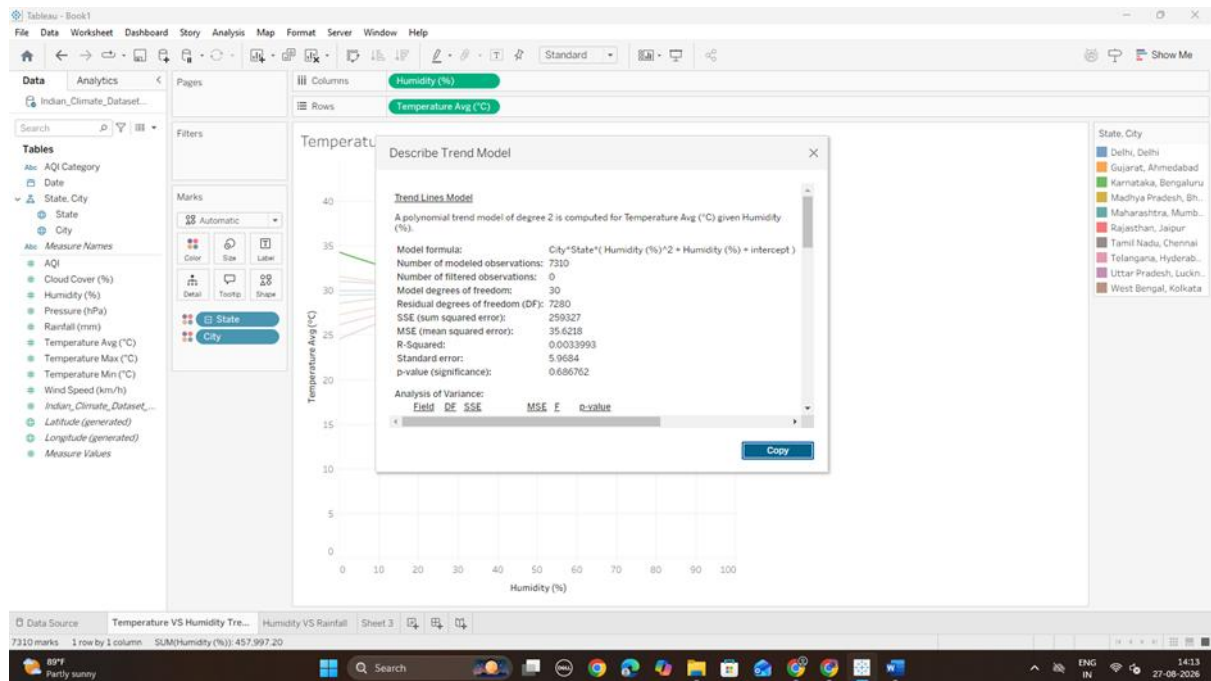
Step 4

Right-click again and select Describe Trend Model.

Step 5

Review detailed statistical information such as:

- ANOVA table
- Degrees of Freedom
- Sum of Squared Errors (SSE)
- F-statistic



C. Perform Clustering on the Dataset

1. Create Climate & AQI Clusters

Step 1

Open a new worksheet.

Step 2

Rename it Climate & AQI Clusters.

Step 3

Drag:

- AQI → Columns
- Temperature_Avg (°C) → Rows

Step 4

Go to Analysis and uncheck Aggregate Measures.

Step 5

Drag City to Detail on the Marks card.

2. Apply Clustering

Step 1

Open the Analytics pane.

Step 2

Under Model, drag Cluster onto the chart.

Step 3

In the Clusters dialog box, add these variables:

- AQI
- Temperature_Avg (°C)
- Humidity (%)
- Wind_Speed (km/h)

Step 4

Under Number of Clusters, select:

- Automatic, or
- Enter 3 or 4 clusters manually.

Step 5

Close the dialog box.

Tableau assigns each data point to a cluster and displays them with different colors.

3. Analyze Clusters

Step 1

Right-click the Clusters field.

Step 2

Select Describe Clusters.

Step 3

Check the following tabs:

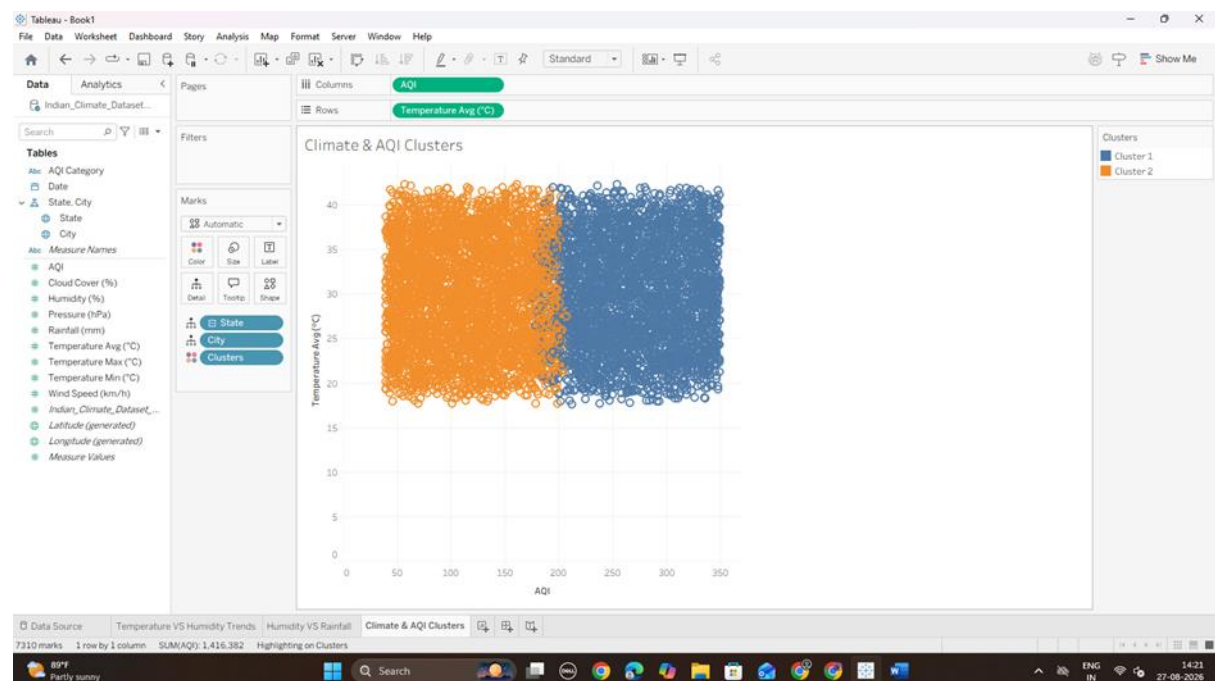
Summary

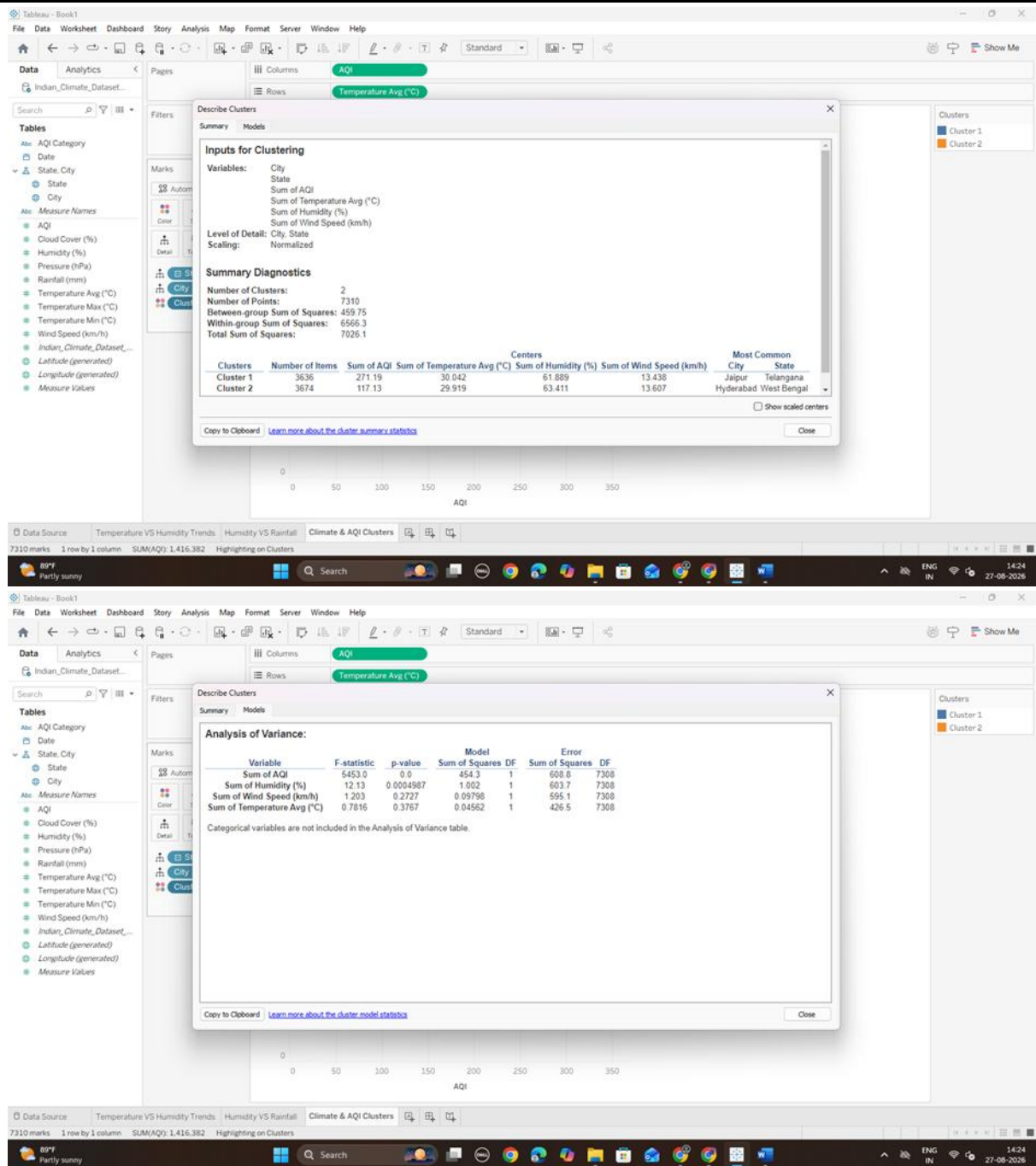
- Mean AQI
- Mean Temperature
- Mean Humidity
- Mean Wind Speed

Models

- ANOVA statistics
- F-statistics
- P-values

Result





The data is automatically grouped into similar climate profiles using Tableau's clustering process.

D. Analyze Distributions

1. Create Temperature Distribution Box Plot

Step 1

Create a new worksheet.

Step 2

Rename it Temperature Distributions.

Step 3

Drag:

- City → Columns
- Temperature_Max (°C) → Rows

Step 4

Change the measure from SUM if necessary, or uncheck Aggregate Measures.

Step 5

Open the Analytics pane.

Step 6

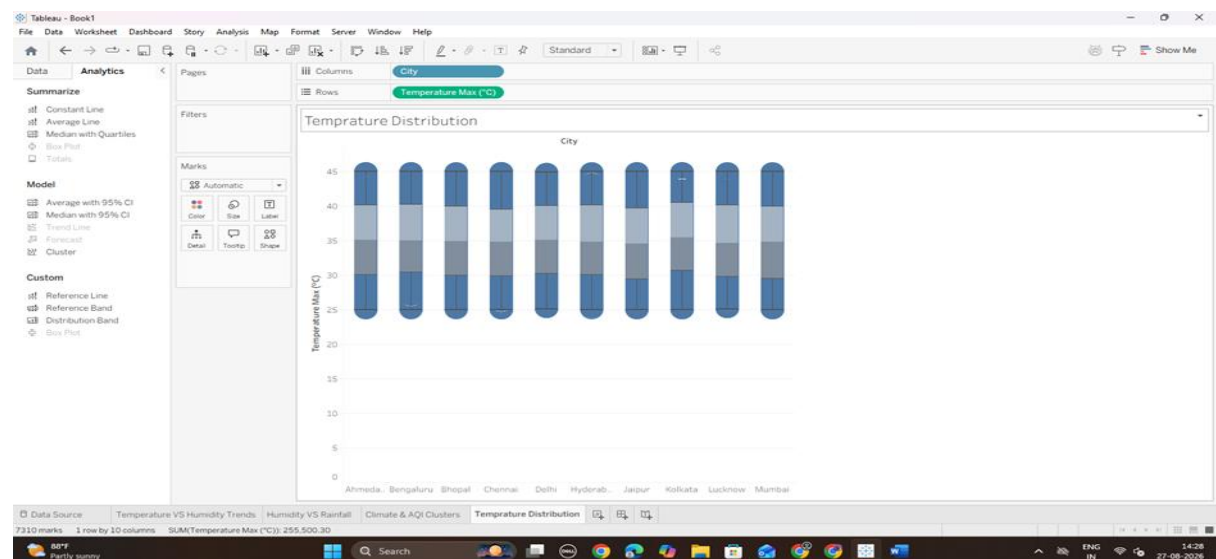
Under Custom, drag Box Plot onto the chart.

Step 7

Observe:

- Maximum / Upper Whisker
- Q3 / Upper Hinge
- Median / Q2
- Q1 / Lower Hinge
- Outliers

Result



The box plot shows the temperature distribution and outliers for each city.

2. Create AQI Histogram

Step 1

Open a new worksheet.

Step 2

Rename it AQI Histogram.

Step 3

Select AQI from the Data pane.

Step 4

Open the Show Me panel.

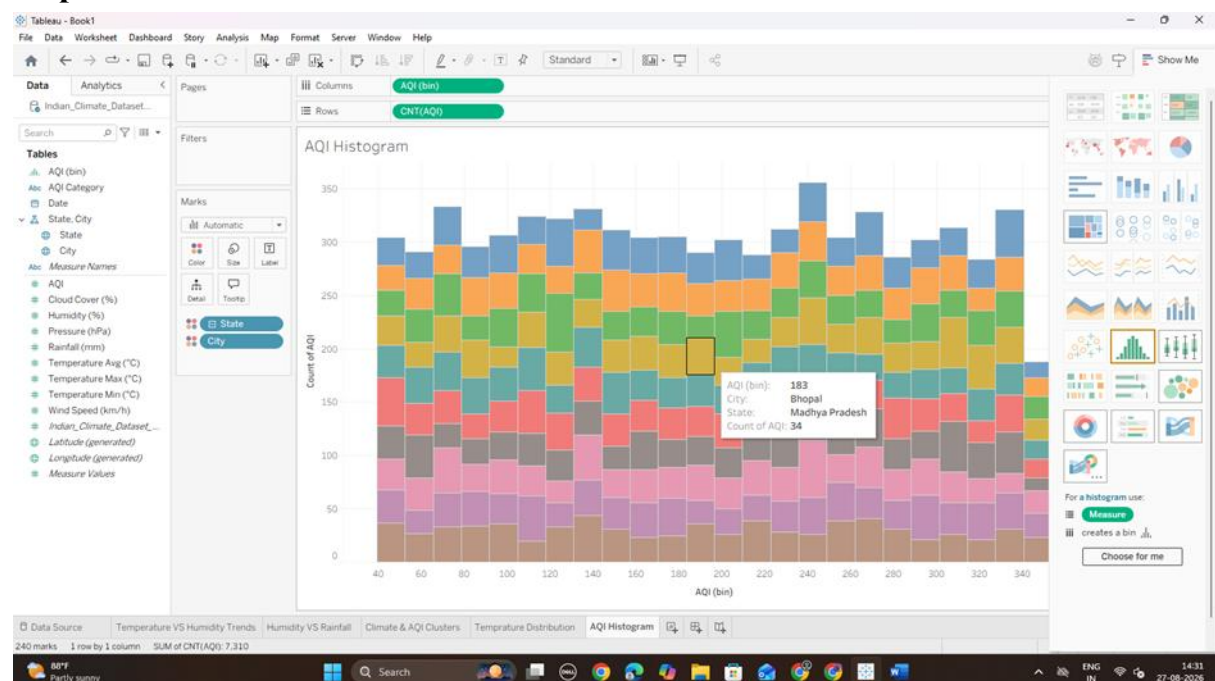
Step 5

Select the Histogram chart.

Tableau automatically creates:

- AQI (bin) on Columns
- CNT(AQI) on Rows

Step 6



Drag City to Color to compare AQI distributions across cities.

E. Apply Forecasting Techniques

1. Create Temperature Forecast Chart

Step 1

Open a new worksheet.

Step 2

Rename it Temperature Forecast.

Step 3

Drag Date to Columns.

Step 4

Change the Date format to Continuous Month.

Step 5

Drag Temperature_Max (°C) to Rows.

Step 6

Ensure it is set to AVG(Temperature_Max).

Step 7

Drag City to Filters.

Step 8

Select 2 or 3 cities, such as:

- Mumbai
 - Delhi
 - Bengaluru
-

2. Enable Forecasting

Step 1

Open the Analytics pane.

Step 2

Under Model, drag Forecast onto the chart.

Step 3

Tableau extends the chart into the future and displays a prediction interval.

3. Customize Forecast

Step 1

Right-click the chart.

Step 2

Select Forecast → Forecast Options.

Step 3

Set:

- Forecast Length: Exactly 6 Months
- Prediction Interval: 95%

Step 4

To check forecast quality:

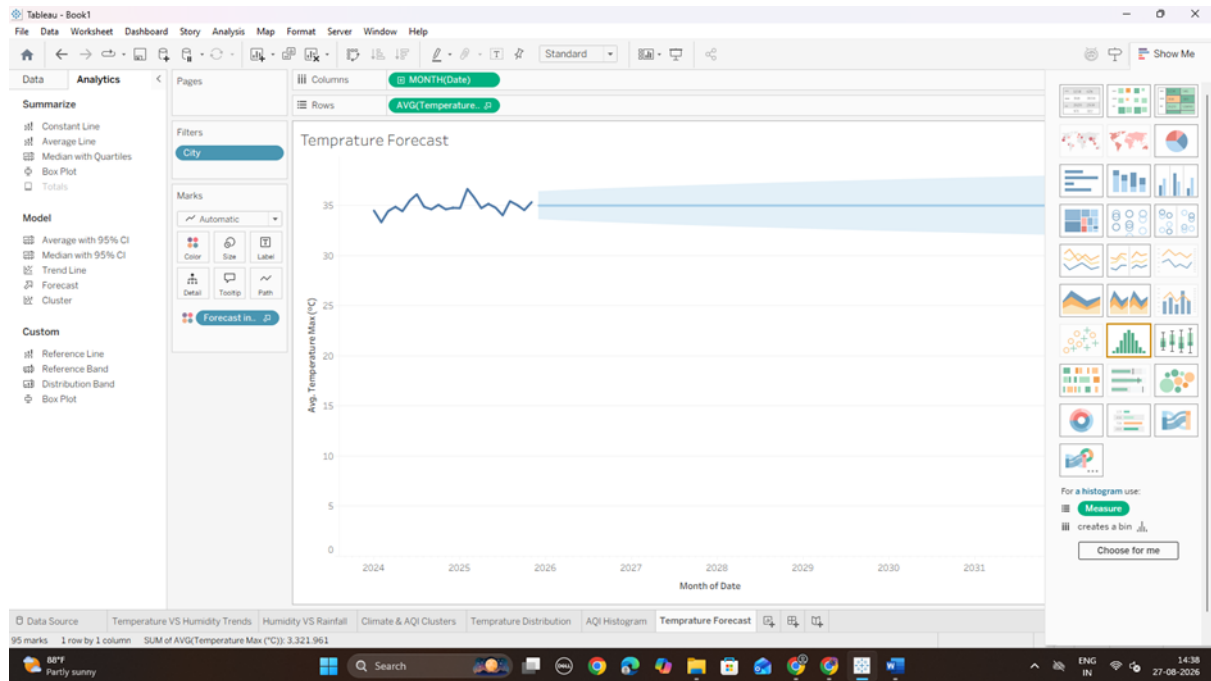
- Right-click the chart.
- Select Forecast → Describe Forecast.

Step 5

Review:

- RMSE
- Additive or Multiplicative model
- Trend
- Seasonality

Result



A six-month temperature forecast is generated based on historical climate data.

F. Compare Different Statistical Models

1. Create Model Comparison Worksheet

Step 1

Create a new worksheet.

Step 2

Rename it Model Comparison.

Step 3

Drag Temperature_Max (°C) to Columns.

Step 4

Set it as a Continuous Dimension or AVG.

Step 5

Drag AQI to Rows.

Step 6

Set it as a Continuous Dimension or AVG.

Step 7

Drag City to Detail on the Marks card.

2. Compare Models

Step 1

Open the Analytics pane.

Step 2

Add a Linear Trend Line.

Step 3

Hover over the line and note:

- R^2
- P-value

Step 4

Right-click the trend line and select Edit Trend Lines.

Step 5

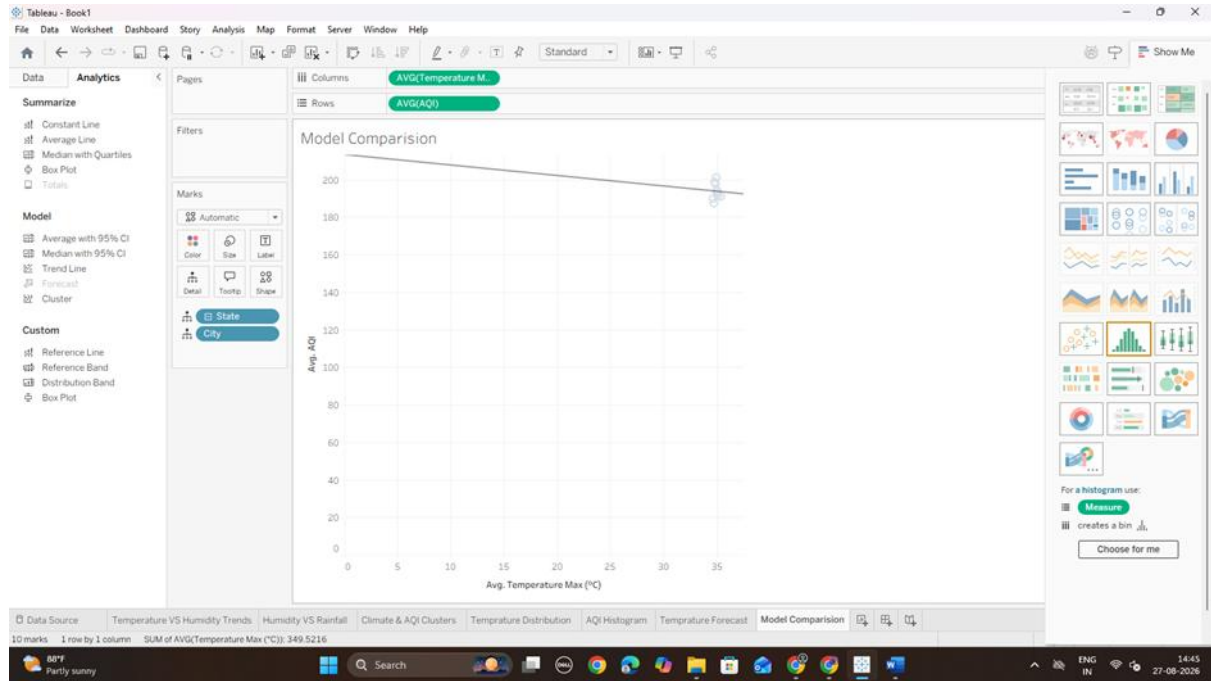
Compare:

- Linear
- Logarithmic
- Exponential
- Polynomial Degree 2
- Polynomial Degree 3

Step 6

Compare the R^2 values.

According to the practical instructions, the model with a higher R^2 and a p-value below 0.05 is considered the best fit.



G. Export and Interpret Statistical Results

1. Export Results

Step 1

Open the Model Comparison worksheet or any worksheet containing a trend line or forecast.

Step 2

Right-click the trend line or chart.

Step 3

Select:

- Describe Trend Model, or
- Describe Forecast

Step 4

In the pop-up window, click:

- Copy, or
- Export

This provides the statistical summary and parameter estimates.

2. Interpret the Results

Document these important findings:

1. Equation Parameters

Check the:

- Intercept
- Slope
- Formula coefficients

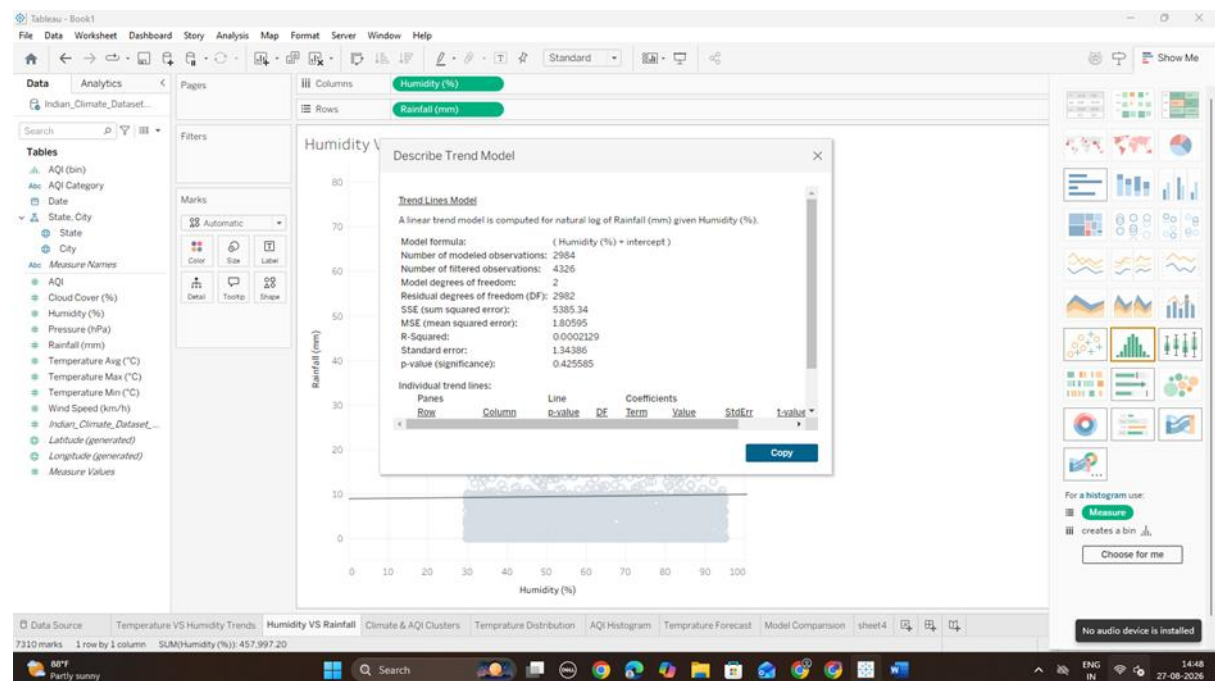
2. R² Value

R² shows how much variation in the data is explained by the model.

Example: $R^2 = 0.74$ means that 74% of the variance is explained by the relationship in the model.

3. P-Value

A small p-value indicates that the relationship is statistically significant.



Trend Lines Model

A linear trend model is computed for natural log of Rainfall (mm) given Humidity (%).

Model formula: (Humidity (%) + intercept)

Number of modeled observations: 2984

Number of filtered observations: 4326

Model degrees of freedom: 2

Residual degrees of freedom (DF): 2982

SSE (sum squared error): 5385.34

MSE (mean squared error): 1.80595

R-Squared: 0.0002129

Standard error: 1.34386

p-value (significance): 0.425585

Individual trend lines:

Panes Line Coefficients

Row	Column	p-value	DF	Term	Value	StdErr	t-value	p-value
Rainfall (mm)	Humidity (%)	0.425585	2982	Humidity (%)	0.0010491	0.0013166		
	intercept	2.18398	0.0854115	25.5701	< 0.0001			

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